

AE-375: Stock solutions of Na₂PO₄ and Na₂CO₃ in H₂O and KOH in D₂O

Date: 2024-11-11
Tags: Stocksolution AE
Category: Two-photon
Status: Done
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Reaction scheme/sample structure

Chemical Formula: HKO
Molecular Weight: 56,10530

Na ₂ HPO ₄	Na ₂ CO ₃	KOH
Chemical Formula: HNa ₂ O ₄ P	Chemical Formula: CNa ₂ O ₃	
Molecular Weight: 141,95730	Molecular Weight: 105,98754	

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	AE-243: 1 M Stocksolution of KOH in water

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]
Na ₂ HPO ₄	7558-79-4	/	0.5	/	70.98	71.27	141.96	/	/
Na ₂ CO ₃	497-19-8	/	0.5	/	53.00	52.99	105.99	/	/
KOH	1310-58-3	/	0.5	/	28.06	21.81	56.11	/	/
Milli-Q water	7732-18-5	/	/	/	/	/	18.02	/	2 * 5 mL
D ₂ O	7789-20-0	C119864	/	/	/	/	20.03	/	3.886 mL

Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#), unless other specified.

Date	Time	Step	Observations
		Done under air	
11.11	10:05	Each in a separate 10 mL vial Na ₂ HPO ₄ , Na ₂ CO ₃ and KOH were added	-1: Na ₂ HPO ₄ , -2: Na ₂ CO ₃ , -3: KOH

	10:40	To -1 and -2 each water (5 mL) was added. To -3 D2O (3.9 mL).	For each sample a 0.1 M mixture was obtained
	10:45	Each vial was shortly vortexed (Equipment - VWR® VV3, Vortex Mixer , stage 4/6), till everything dissolved	
	10:50	The pH value was determined for each sample using pH paper from 7.5 to 14	AE-375-1, -2, -3
		From now on under inert conditions	
		3 10 mL Schlenk flask were prepared	
		Each sample was transferred into a 10 mL Schlenk flask in argon counterflow using glas pipettes	
	15:25 - 15:45	AE-375-1 was degassed by bubbling with argon and subsequently stored at rt under argon	degassing.jpg
	15:50 - 16:15	AE-375-2 was degassed by bubbling with argon and subsequently stored at rt under argon	
	16:17 - 16:40	AE-375-3 was degassed by bubbling with argon and subsequently stored at rt under argon	

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
11.11	10:52	AE-375-1	pH	pH paper (7.5 - 14)	/	pH 1.jpg	pH 1.jpg	/	pH = 9
11.11	10:52	AE-375-2	pH	pH paper (7.5 - 14)	/	pH 1.jpg pH 2.jpg	pH 1.jpg pH 2.jpg	/	pH = 10.5
11.11	10:52	AE-375-3	pH	pH paper (7.5 - 14)	/	pH 2.jpg	pH 2.jpg	/	pH = 13 to 14

Results

For each sample degassed 0.1 M solution were obtained

Linked experiment

- [AE-243: 1 M Stocksolution of KOH in water](#)

Linked resource

Equipment - [VWR® VV3, Vortex Mixer](#)

Attached files

AE-375.png

sha256: e6a577d3a5eee53dc5601322959c19a33837bd3b3046bc26abb38bacfbbe3d72



AE-375.cdxml

sha256: d60cb63d4d5d8299bd2e870757d0ce5daab2036e1fa8dccdf02bc0cab9998043

pH-2.jpg

sha256: 9ea2204ec0b3b7eac1913305113dae9a3371b7045ad934be6c1b690c7da1fadf



degassing.jpg

sha256: 7a078d4217a91592e3e0babd58f998f5254644970296f109398ed5279a867b93



pH-1.jpg

sha256: a0e4f73228d0291c9aec4d2a312c0e887bc7fb11f961785bfa58a9f7176b04c5



Unique eLabID: 20241111-95e35574375b3b95937e08340dead58036b37f27

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1489>